

Hypercycles and dissimilarities

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Abstract:

This paper discusses the extension of interval hypergraphs to hypercycles and their connection with dissimilarities. After recalling a polynomial algorithm to recognize if a given hypergraph is a hypercycle, we characterize dissimilarities whose ball, 2-ball and cluster hypergraphs are hypercycles, discuss their links with robinsonian and circular anti-robinsonian dissimilarities and provide two algorithms to approximate a given dissimilarity by a dissimilarity whose cluster hypergraph is a hypercycle.

Keywords:

Clustering system, dissimilarity, cycle, interval hypergraph, hypercycle.

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1 Introduction

In many usual models of classification on a set X , the clusters are connected subsets of a path. This is the case for partitions, hierarchies, and more generally for pyramids. Those models are related to *interval hypergraphs* (cf. for instance Diday, 1986 and Fichet, 1986).

Those models are well suited when the data follow a linear order. For instance Robinson (1951) used interval hypergraphs (he more precisely a particular kind of distances that are now called *robinsonian dissimilarities*. See part 2) to find the chronological order between archeological data.

Even though most of the data are not related with a linear order, pyramids are a popular tool in data analysis because of its graphical representation (figure 6) which generalize the hierarchical one (known as dendrogram). It's indeed a convenient way to represent overlapping clusters without losing readability (which is often the case when dealing with models admitting overlapping clusters).

Nevertheless, since all the elements are not equivalent: the first and the last element of the order plays a particular role as the only one cluster which these two elements is the whole set, using pyramids with data that are not linked with a linear order may lead to misinterpretation (see part 8). Moreover, some data set are linked with a cyclic order (a year for instance) and using a linear order for them is irrelevant. See for instance Hubert, Arabie and Meulman (1997, 1998) for examples in psychology.

Using hypercycles instead of interval order can be a solution. Indeed, they are a generalization of pyramids (thus may help in finding the real order when dealing with data following a linear order), in hypercycles all the elements plays the same role, they are linked with a circular order and they admit a graphical representation (figure 5) as clear as pyramids.

Last but not the least they have good theoretical properties as shown by Quillot (1984) or Hsu and McConnel (2003) for instance.

We will in this paper study the use of hypercycles in numerical taxonomy (that is data are described by a distance).

After recalling some definition related with numerical taxonomy, clusters and dissimilarities, we will show how define dissimilarities which clusters are hypercycles (part 2). We will then focus on hypercycles and their links their links to interval hypergraphs. In particular, we will show that one can define one structure with the other (and conversely) and recall a linear algorithm which recognize if a given hypergraph is a hypercycle.

In order to produce hypercycles in numerical taxonomy we then show the dissimilarity model associated with hypercycles and show that all the hypercycle are not in relation with a dissimilarity.

After that, we propose an algorithm in $\mathcal{O}(|X|^3)$ time and $\mathcal{O}(|X|^2)$ space to recognize if a given dissimilarity is a circular one. This algorithm can also be used to recognize if a given dissimilarity is a robinsonian one (thus an interval order). Moreover, it has the same complexity of the quickest algorithm – as our best knowledge – designed to check whether a given dissimilarity is a robinsonian one or not (due to Chepoi and Fichet, 1997). We then use these characterization to give two algorithms to approximate a given dissimilarity by a circular one. The first algorithm needs an underlying vertex cycle to be given and the second one tends to find a “good” one.

We conclude after a section devoted to an example.

2 Mathematical material and first definitions

Let X be a finite set with n elements. A finite *hypergraph* $\mathcal{H} = (X, \mathcal{E})$ is defined by a set X and a family $\mathcal{E}$ of nonempty subsets (called *edges*) of X (whose elements are called *vertices*). A *clustering system* $\mathcal{K} = (X, \mathcal{E})$ is a hypergraph admitting as edges X and $\{x\}$ for all $x \in X$. A hypergraph $G = (X, E)$ whose edges are pairs of elements is simply called *graph*, and in this case elements of E will be noted xy instead of $\{x, y\}$ for simplicity. Given a graph $G = (X, E)$, a *path* between two elements x and y in G is a list $x = x_0, x_1, \dots, x_p = y$ such that for all $0 \leq i \leq p-1$, $x_i x_{i+1} \in E$ and $x_i \neq x_j$ for all $i \neq j$. By extension, a *path* is a graph $P = (X, E)$ where there is an ordering of $X = \{x_1, x_2, \dots, x_n\}$ and where $E = \{x_i x_{i+1} \mid 1 \leq i \leq n-1\}$. A cycle $C = (X, E)$ is a graph where there is an ordering of $X = \{x_1, x_2, \dots, x_n\}$ and where $E = \{x_i x_{i+1} \mid 1 \leq i \leq n-1\} \cup \{x_1 x_n\}$.

Given a graph $G = (X, E)$, a subset A of X is said to be a *connected class* of G if for all x and y in A there is a path $x, x_1, \dots, x_{p-1}, y$ between x and y in G such that $x_i \in A$ for all $0 \leq i \leq p$. If X is a connected class of G , G is said to be *connected*. Let $G = (X, E)$ be a graph and $\mathcal{H} = (X, \mathcal{E})$ a hypergraph such that each edge of $\mathcal{H}$ is a connected class of G . G is said to be an *underlying vertex graph* of $\mathcal{H}$ and $\mathcal{H}$ is said to be *rigid* on G (for more details about rigidity, see for instance Flament, Degenne and Verges, 1979 or Osswald, 2003).

These definitions allow us to define an interval hypergraph. $\mathcal{H} = (X, \mathcal{E})$ is an *interval hypergraph* if and only if $\mathcal{H}$ is a hypergraph rigid on a path. Besides the combinatorial properties of interval hypergraphs (see for instance Berge, 1987 or Duchet, 1995), one of the main interests of those hypergraphs in classification is the relationship between interval hypergraphs having X and $\{x\}$ for all $x \in X$ as edges and *dissimilarities* whose “classes” are interval hypergraphs.

A *dissimilarity* d on X is a function from $X \times X$ to the nonnegative real numbers which is symmetrical ($d(x, y) = d(y, x)$ for $x, y \in X$) and has a zero-diagonal ($d(x, x) = 0$ for all $x \in X$). Moreover, a dissimilarity d on X is said to be *proper* if $d(x, y) = 0$ is equivalent to $x = y$. Given a dissimilarity d , we associate to any subset A of X its *diameter* defined by $\text{diam}_d(A) = \max\{d(x, y) \mid x, y \in A\}$. In this paper we focus on proper dissimilarities. Dissimilarity and proper dissimilarity are considered as synonyms in the rest of the text.

They are many ways to associate “classes” to a dissimilarity. In numerical taxonomy, we usually follow Jardine and Sibson’s 1971 approach, and associate to each dissimilarity d its *cluster hypergraph* $\mathcal{H}_d = (X, \mathcal{K}_d)$ whose edges are exactly the *maximal cliques* of *threshold graphs* of d . A threshold graph at level α of a dissimilarity d on X is the graph $G_\alpha = (X, E_\alpha)$ such that

$xy \in E_\alpha$ if and only if $d(x, y) \leq \alpha$. A clique A of a graph $G = (X, E)$ is a subset of X such that $xy \in E$ for all $x, y \in A$ and a maximal clique of G is a clique maximal with respect to set inclusion.

Nevertheless, there can be an exponential number of clusters for a given dissimilarity d on X . So, we usually associate to d either its *ball hypergraph*, or its *2-ball hypergraph* which has $\mathcal{O}(n^2)$ edges. A *ball* $B(x, \lambda)$ contains all the elements z such that $d(x, z) \leq \lambda$; the hypergraph composed of the balls of d is noted $\mathcal{B}_d$. A *2-ball* $B(x, y)$ contains all the elements z such that $\max\{d(x, z), d(y, z)\} \leq d(x, y)$; the hypergraph composed of the 2-balls of d is noted $2\text{-}\mathcal{B}_d$.

More generally, extending Bertrand and Janowitz, 2003, one can associate to each dissimilarity d on X and each subset $Y \subseteq X$ with $|Y| \leq k$ a k -ball $B(Y) = \bigcap_{x \in Y} B(x, \text{diam}(Y))$. The hypergraph composed of the k -balls of d is noted $k\text{-}\mathcal{B}_d$. Note that 1-balls (which are the singletons) do not coincide with balls and that, unlike Bertrand and Janowitz 2003, we do not suppose that $|Y| = k$. One trivial property of k -balls is that $(k-1)\text{-}\mathcal{B}_d \subseteq k\text{-}\mathcal{B}_d$ and that $k\text{-}\mathcal{B}_d$ has $\mathcal{O}(n^k)$ elements. 3-balls will have a particular role in the section devoted to circular dissimilarities.

One can prove that the sets of dissimilarities whose cluster, ball, or 2-ball hypergraphs are interval hypergraphs coincide (and more generally hypergraphs that admit an underlying vertex tree, see for instance Brucker, 2005). The dissimilarities in this set are called *robinsonian dissimilarities* (Fichet, 1986; Durand and Fichet, 1988). Formally speaking, a robinsonian dissimilarity d on X , is a dissimilarity for which there exists a linear order $<$ on X , called *compatible order*, such that for all $x, y, z \in X$ with $x < y < z$ we have $\max\{d(x, y), d(y, z)\} \leq d(x, z)$. Considering an order $i \leq j$ if and only if $x_i < x_j$, this condition is equivalent to: for all $1 \leq i < j \leq n-1$, $d(x_i, x_j) \leq d(x_i, x_{j+1})$ and for all $2 \leq i < j \leq n$, $d(x_i, x_j) \leq d(x_{i-1}, x_j)$.

Since each interval hypergraph containing X and $\{x\}$ (for all $x \in X$) as edges is a cluster hypergraph of a robinsonian dissimilarity, the two notions of interval graphs and robinsonian dissimilarities are closely related through cluster hypergraphs.

A natural generalization of those structures is to consider a “circular order” instead of a linear one. Two approaches are possible, generalizing either robinsonian dissimilarities or interval hypergraphs.

Hubert, Arabie and Meulman, 1997 and 1998 choose the first approach and generalize robinsonian dissimilarities to *circular anti-robinsonian dissimilarities*. Using their own characterization, a dissimilarity d on X is a circular anti-robinsonian dissimilarity if there exists a linear order $<$ on X (we label the elements of X as $x_1 < x_2 < \dots < x_n$) such that the four hereunder conditions are satisfied for $1 \leq i \leq n-2$ and $i+1 < j \leq n-1$:

$$\text{if } d(x_{i+1}, x_j) \leq d(x_i, x_{j+1}) \quad \text{then} \quad d(x_{i+1}, x_j) \leq d(x_i, x_j) \quad (1)$$

$$\text{and } d(x_{i+1}, x_j) \leq d(x_{i+1}, x_{j+1})$$

$$\text{if } d(x_{i+1}, x_j) \geq d(x_i, x_{j+1}) \quad \text{then} \quad d(x_i, x_j) \geq d(x_i, x_{j+1}) \quad (2)$$

$$\text{and } d(x_{i+1}, x_{j+1}) \geq d(x_i, x_{j+1})$$

$$\text{if } d(x_{i+1}, x_n) \leq d(x_i, x_1) \quad \text{then} \quad d(x_{i+1}, x_n) \leq d(x_i, x_n) \quad (3)$$

$$\text{and } d(x_{i+1}, x_n) \leq d(x_{i+1}, x_1)$$

$$\text{if } d(x_{i+1}, x_n) \geq d(x_i, x_1) \quad \text{then} \quad d(x_i, x_n) \geq d(x_i, x_1) \quad (4)$$

$$\text{and } d(x_{i+1}, x_1) \geq d(x_i, x_1)$$

It is clear that a robinsonian dissimilarity is a circular anti-robinsonian dissimilarity.

Quillot, 1984, chooses the second approach and generalizes interval hypergraphs to *hypercycles*. As interval hypergraph are related to paths, hypercycles are related to *cycles*. By definition, a hypercycle is simply a hypergraph $\mathcal{H} = (X, \mathcal{E})$ rigid on a cycle.

These two approaches are not equivalent, because neither the cluster hypergraph, the 2-ball hypergraph nor the ball-hypergraph of a circular anti-robinsonian dissimilarity is a hypercycle in general. To preserve the relationship between dissimilarities and hypergraphs we propose in this article two new kinds of dissimilarities called *precircular* and *circular dissimilarities*. We also show that, unlike robinsonian dissimilarities, the sets of dissimilarities whose 2-ball and cluster hypergraphs are hypercycles do not coincide and we characterize them.

3 Hypercycles

This short section recalls some properties of hypercycles, in particular their links with interval hypergraphs and an algorithm dedicated to their recognition.

3.1 Hypercycles and interval hypergraphs

It is clear that any interval hypergraph $\mathcal{H} = (X, \mathcal{E})$ with underlying vertex path $P = (X, \{x_1x_2, \dots, x_{n-1}x_n\})$ is also a hypercycle with underlying vertex cycle $C = (X, \{x_1x_2, \dots, x_{n-1}x_n, x_nx_1\})$. However, interval hypergraph $\mathcal{H} = (X, \mathcal{E})$ where there are two edges A and B for which $A \cup B = X$ may have an underlying vertex cycle which does not induce an underlying vertex path by deleting an edge, as shown in figure 1.

Despite this fact, it is possible to characterize interval hypergraphs using hypercycles by adding a “fake” element, as shown in proposition 1.

Proposition 1. *A hypergraph $\mathcal{H} = (X, \mathcal{E})$ is an interval hypergraph if and only if $\mathcal{H}' = (X \cup \{z\}, \mathcal{E})$ is a hypercycle for $z \notin X$.*

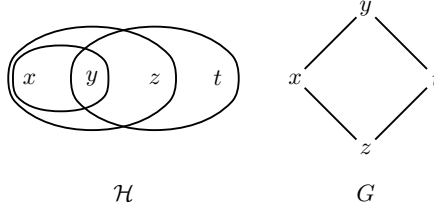


Figure 1: An interval hypergraph $\mathcal{H} = (X, \{\{x, y\}, \{x, y, z\}, \{y, z, t\}\})$ and one of its underlying vertex cycle $G = (X, \{xy, yt, tz, zx\})$ which does not contain an underlying vertex path

Proof. If $\mathcal{H}$ is an interval hypergraph, it is clear that $\mathcal{H}'$ is also an interval hypergraph, thus a hypercycle.

Conversely, let $\mathcal{H}'$ be a hypercycle, and $C = (X, E)$ be one of its underlying vertex cycles. We can assume without loss of generality that this cycle is such that $C = (X, \{zx_1, x_1x_2, \dots, x_{n-1}x_n, x_nz\})$. Let A be an edge of $\mathcal{H}'$. Since $z \notin A$, A is an interval of the linear order $<$ defined such that $x_1 < x_2 < \dots < x_n < z$. Thus $\mathcal{H}$ is an interval hypergraph $\square$

Conversely, it is also possible to define hypercycles using interval hypergraphs as shown in proposition 2 proven in Quillot 1984. To do this, we have to introduce the *anti-Helly property* (which refers to the Helly property introduced by Duchet, among others, to characterize hypergraphs rigid on a tree).

Definition 1. Let $\mathcal{H} = (X, \mathcal{E})$ be a hypergraph. $\mathcal{H}$ is said to be *anti-Helly* if for any of its edges A_1, A_2 and A_3 , if $A_1 \cap A_2 \cap A_3 = \emptyset$ there exists $i, j, k \in \{1, 2, 3\}$, $i \neq j$ and $i \neq k$, such that $A_i \subseteq A_j \cup A_k$.

Proposition 2 (Quillot 1984). Let $\mathcal{H} = (X, \mathcal{E})$ be a hypergraph, and define $\mathcal{H}' = (X, \mathcal{E}')$ such that:

- $\mathcal{E} \subseteq \mathcal{E}'$
- $X \in \mathcal{E}'$ and for all $x \in X$, $\{x\} \in \mathcal{E}'$ and $X \setminus \{x\} \in \mathcal{E}'$,
- if $A \in \mathcal{E}'$ then $X \setminus A \in \mathcal{E}'$
- if $A_1, A_2 \in \mathcal{E}'$ and $A_1 \cup A_2 \neq X$ then $A_1 \cap A_2 \in \mathcal{E}'$.

$\mathcal{H}$ is a hypercycle if and only if $\mathcal{H}'$ has the anti-Helly property and there exists $x \in X$ such that the edges of $\mathcal{H}$ which do not contain x form an interval hypergraph.

Proposition 2 gives us a way to check whether a hypergraph is a hypercycle or not. Nevertheless, since a hypercycle $\mathcal{H} = (X, \mathcal{E})$ has at most $\mathcal{O}(n^2)$ edges, computing $\mathcal{H}'$ takes $\mathcal{O}(n^5)$ operations and testing the anti-Helly property $\mathcal{O}(n^7)$, as there are $\mathcal{O}(n^6)$ different subsets with three elements. This mathematical property of hypercycles does not lead to an efficient recognition algorithm. The next section will give a linear algorithm, that is in $\mathcal{O}(|X| * |\mathcal{E}|)$ operations, to check whether a hypergraph is a hypercycle or not.

3.2 Recognition of hypercycles

Hsu and McConnell (2003) have shown that a structure, called *PC-tree*, which is a generalization of well-known PQ-trees (Booth and Lueker, 1976) can be successfully used to recognize whether a given hypergraph is a hypercycle or not. They show that one can check whether a hypergraph $\mathcal{H} = (X, \mathcal{E})$ is a hypercycle or not in $\mathcal{O}(|X| * |\mathcal{E}|)$ operations. We will not give further formal definition of PC-trees or the way to construct them; for more information about them, see Hsu and McConnell (2003) for instance.

4 Hypercycles and dissimilarities

This section is devoted to dissimilarities. We characterize here those whose cluster hypergraphs or k -ball hypergraphs are hypercycles. To do this we will use linear orders and associate them to cycles. To any linear order $<$ on X (labelling the elements of X such that $x_1 < \dots < x_n$) we define a cycle $C_{<} = (X, E)$ where $E = \{x_1x_2, x_2x_3, \dots, x_{n-1}x_n, x_nx_1\}$. We denote by $<^{-1}$ the linear order on X such that $x_n <^{-1} x_{n-1} <^{-1} \dots <^{-1} x_1$ and by $<_{x_i}$ the linear order on X such that $x_i <_{x_i} x_{i+1} <_{x_i} \dots <_{x_i} x_n <_{x_i} x_1 <_{x_i} \dots <_{x_i} x_{i-1}$, it is clear that $C_{<^{-1}}$ and $C_{<_{x_i}}$ are equal to $C_{<}$.

Associating linear orders to cycles allows us to say that a linear order $<$ is *circularly compatible* with a hypergraph $\mathcal{H}$ if and only if $\mathcal{H}$ is a hypercycle admitting $C_{<}$ as underlying vertex cycle.

4.1 Cluster properties

Despite the fact that hypercycles are closely related to interval hypergraphs (*cf.* section 3.1), the small changes between these two structures have a huge impact when dealing with dissimilarities.

First of all there exists hypercycles which cannot be cluster hypergraphs of some dissimilarities (figure 2 is such a hypergraph because if we label A , B and C the three clusters, $(A \cap B) \cup (A \cap C) \cap (B \cap B)$ is the whole set and thus no index will satisfy the conditions of theorem 1) whereas any interval hypergraph is a cluster hypergraph of a dissimilarity.

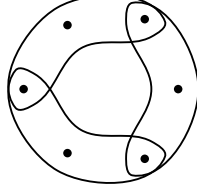


Figure 2: A hypercycle which is not a cluster hypergraph of some dissimilarity

Moreover, it is well known that for a given dissimilarity d on X the following three propositions are equivalent, for any $k \geq 2$:

- the cluster hypergraph of d is an interval hypergraph,
- the k -ball hypergraph of d is an interval hypergraph,
- the ball hypergraph of d is an interval hypergraph.

These properties no longer hold for hypercycles, since the 2-ball hypergraph of the dissimilarity d_1 in table 1 is a hypercycle but not its cluster hypergraph, and the ball hypergraph of dissimilarity d_2 in table 1 is a hypercycle but neither its cluster hypergraph nor its 2-ball hypergraph are hypercycles.

Table 1: dissimilarities d_1 and d_2

$d_1 :$	x	0				
	y	1	0			
	z	3	2	0		
	t	2	2	2	0	
	u	2	3	3	2	0
		x	y	z	t	u

$d_2 :$	x	0				
	y	2	0			
	z	3	2	0		
	t	2	2	1	0	
	u	2	3	3	1	0
		x	y	z	t	u

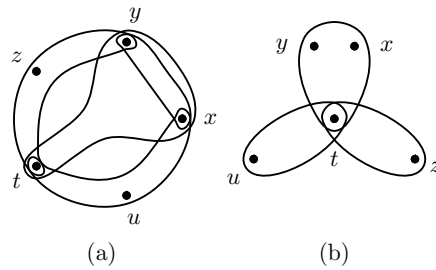


Figure 3: clusters of d_1 (a) and 2-balls of d_2 (b) (table 1)

Nevertheless, proposition 3 shows the remaining relationship between clusters, k -balls and balls in a hypercycle context.

Proposition 3. *Let d be a dissimilarity on X . We have the following implications, for any $k \geq 2$:*

*if the cluster hypergraph of d is a hypercycle rigid on cycle C then
the k -ball hypergraph of d is a hypercycle rigid on cycle C which then implies that
the ball hypergraph of d is a hypercycle rigid on cycle C .*

Proof. Let d be a dissimilarity on X and C a cycle such that the clusters of d are connected classes of C . Let $Y \subseteq X$. Let A be a clique of d included in $B(Y)$ and such that $\text{diam}(A) = \text{diam}(Y)$. There exists a maximal clique A' (thus a connected class of C) including Y and thus included in $B(Y)$. One can conclude that:

- for all $x, y \in Y$, there exists a path from x to y in A' which is a connected class of C , and therefore in $B(Y)$.
- for all $x \in B(Y)$, there exists a maximal clique A_x of diameter $\text{diam}(Y)$ such that $Y \subseteq A_x$ and $x \in A_x$: we have $A_x \subseteq B(Y)$. There exists a path from x to any $y \in Y$ in A_x which is a connected class of C and therefore in $B(x)$.

Thus, for all $x, y \in B(Y)$ there is a path from x to $z \in Y$ and from y to $t \in Y$ in $B(Y)$. Since for all $z, t \in Y$ there is a path from z to t in $B(Y)$, there is a path from x to y in $B(Y)$, which is a connected class of C .

Since k and Y have been arbitrarily chosen, one can conclude that:

If the cluster hypergraph of d is a hypercycle rigid on cycle C then
the k -ball hypergraph of d is a hypercycle rigid on cycle C .

Consider now a dissimilarity d such that its 2-ball hypergraph is a hypercycle rigid on cycle C , and consider a ball $B(x, \alpha)$. Let $y \in B(x, \alpha)$; since $B(x, y)$ is a connected class of C and $B(x, y) \subset B(x, \alpha)$, there is a path from y to x in $B(x, \alpha)$ which is a connected class of C , thus $B(x, \alpha)$ is a connected class of C $\square$

Proposition 4 refines proposition 3 and shows that there are only three distinct sets of dissimilarities. The set of dissimilarities whose ball hypergraphs are hypercycles, which strictly includes the set of those whose 2-ball hypergraphs are hypercycles, which strictly includes the set of dissimilarities whose cluster hypergraphs are hypercycles (which is equivalent to the fact that the 3-ball hypergraph is a hypercycle).

Proposition 4. *Let d be a dissimilarity on X . The two following propositions are equivalent.*

1. *The cluster hypergraph of d is a hypercycle on cycle C .*
2. *The 3-ball hypergraph of d is a hypercycle on cycle C .*

Proof. According to Proposition 3, we have (1) $\Rightarrow$ (2). To prove that (2) $\Rightarrow$ (1), consider a dissimilarity d on X and cycle C such that the set of 3-balls of d is a hypercycle on C .

Suppose that there exists a cluster A of d such that A is not a connected class of C . Then there exist $x, y \in A$ such that there does not exist a path from x to y of elements of A . Since there exists two disjoint paths from x to y in C , there exists t_1 in the first path and t_2 in the second one such that t_1 and t_2 are not in A . Thus every path from x to y in C contains either t_1 or t_2 .

There exists then $a_1, a_2 \in C$ such that $d(a_1, t_1) > \text{diam}(A)$ and $d(a_2, t_2) > \text{diam}(A)$. Several cases may occur:

- $a_1 = a_2$: in this case, the 3-ball $B(Y)$ where Y contains only a_1, x and y (a_1 may be equal to x or y) is a connected class of C which does not contain t_1 and t_2 . Then a path from x to y in C which has neither t_1 nor t_2 would exist, which is impossible.
- $a_1 \neq a_2$: consider the set U_x containing only a_1, a_2 and x (x may be equal to a_1 or a_2) and the set U_y containing only a_1, a_2 and y (y may be equal to a_1 or a_2). We have then that $B(U_x)$ and $B(U_y)$ are connected classes of C , there exist then a path from x to a_1 in C having only elements of $B(U_x)$ and a path from y to a_1 in C having only elements of $B(U_y)$, thus a path from x to y in C having only elements of $B(U_x) \cup B(U_y)$. Since $t_1, t_2 \notin B(U_x) \cup B(U_y)$, a path from x to y in C which does not have t_1 and t_2 would exist, which is not possible.

Thus our hypothesis that A is not connected does not hold $\square$

4.2 Metric properties

We will now characterize dissimilarities whose k -ball hypergraphs are hypercycles. Since there are only three cases of interest (cf. propositions 3 and 4), we will characterize the dissimilarities whose:

- ball hypergraphs are hypercycles (proposition 5),
- 2-ball hypergraphs are hypercycles (proposition 6),
- 3-ball hypergraphs (thus cluster hypergraphs and k -ball hypergraphs with $k \geq 3$) are hypercycles (proposition 9).

Proposition 5 characterizes dissimilarities whose ball hypergraphs are hypercycles. Equation 5 is a very natural extension of robinsonian dissimilarities condition, taking into account that there are two ways of being connected on a cycle, and that balls $B(x, d(x, z))$ must be connected classes of that cycle (a path for robinsonian dissimilarities, respectively).

Proposition 5. *Let d be a dissimilarity on X and $\leq$ be a linear order on X . The ball hypergraph of d is a hypercycle on $C_{\leq}$ if and only if for all $x \leq_x y \leq_x z \leq_x t$ equation (5) is satisfied*

$$d(x, z) \geq \min\{d(x, y), d(x, t)\} \quad (5)$$

Proof. Let d be a dissimilarity on X and $<$ a linear order on X such that the ball hypergraph of d is a hypercycle on $C_{<}$. Let x, y, z and t be four elements of X such that $x <_x y <_x z <_x t$. Since the ball $B(x, d(x, z))$ is a connected class of $C_{<}$, either y or t are in $B(x, d(x, z))$, thus $d(x, z) \geq \min\{d(x, y), d(x, t)\}$.

Conversely, suppose that d satisfies equation 5 for the linear order $<$. Let $z \in B(x, \alpha)$. If $B(x, \alpha)$ is not equal to X , there exists $t \notin B(x, \alpha)$. Since one can suppose without loss of generality that $z <_x t$, we have, according to equation 5, that $y \in B(x, \alpha)$ for all y such that $x <_x y <_x z$. Thus there exists a path from x to z in $C_{<}$ belonging to $B(x, \alpha)$. Since x, α and z have been arbitrarily chosen, the ball hypergraph of d is a hypercycle on $C_{<}$ $\square$

We will not talk much about the dissimilarities in Proposition 5 because their links with robinsonian, anti-robinsonian and circular dissimilarities cannot be expressed as easily as with *precircular* dissimilarities.

Definition 2. Let $<$ be a linear order on X , d a dissimilarity on X . We will say that a dissimilarity d on X is *precircular* if it exists a linear order $<$ such that for all $x, y, z, t \in X$ for which $x <_x y <_x z <_x t$ equation 6 is satisfied. Such a linear order is said to be *precircularly compatible* with d .

$$d(x, z) \geq \min\{\max\{d(x, y), d(y, z)\}, \max\{d(x, t), d(t, z)\}\} \quad (6)$$

Obviously, a precircular dissimilarity satisfies the equation (5). Proposition 6 proves that precircular dissimilarities are closely related to dissimilarities whose 2-ball hypergraphs are hypercycles.

Proposition 6. Let d be a dissimilarity on X . The 2-ball hypergraph of d is a hypercycle if and only if there exists a linear order $<$ such that:

$$\begin{cases} < \text{ is precircularly compatible with } d, \text{ and} \\ \text{for all } x <_x u <_x y <_x v <_x z, y \in B(x, z) \text{ implies that } \min\{d(u, z), d(v, x)\} \leq d(x, z) \end{cases}$$

Proof. Let d be a dissimilarity on X such that its 2-ball hypergraph is a hypercycle. Then there exist a linear order $<$ such that $C_{<}$ is an underlying vertex cycle of $2-\mathcal{B}$. Let $(x, y, z, t) \in X^4$ such that $x <_x y <_x z <_x t$. The 2-ball $B(x, z)$ is a connected class of $C_{<}$, thus either y or t is in $B(x, z)$ which leads to equation (6): $<$ is precircularly compatible with d . Moreover, for all $(u, v) \in X^2$ such that $x <_x u <_x y <_x v <_x z$ if $y \in B(x, z)$, either u or v is in $B(x, z)$, which leads to $\max\{d(u, z), d(v, x)\} \leq d(x, z)$.

Conversely, suppose that $<$ satisfies the above two conditions. Let x and z be two elements of X . Since $<$ is precircularly compatible with d , we have either that any y such that $x <_x y <_x z$ is in $B(x, z)$ or any y' such that $z <_z y' <_z x$ is in $B(x, z)$. There exists then a path from x to z in $C_{<}$ containing only elements of $B(x, z)$.

Let now be $y \in B(x, z)$. One can suppose without loss of generality that $x <_x y <_x z$. Suppose that there exists v such that $y <_x v <_x z$ such that $v \notin B(x, z)$. Two cases may occur: either $d(x, v) > d(x, z)$ or $d(z, v) > d(x, z)$.

If $d(x, v) > d(x, z)$, for all u such that $x <_x u <_x y$, precircularity, considering the four points $x <_x u <_x y <_x v$ implies that $\max\{d(x, u), d(y, u)\} \leq d(x, z)$. Thus since $d(x, v) > d(x, z)$, the second condition implies that $d(u, z) \leq d(x, z)$. We have then $\max\{d(x, u), d(z, u)\} \leq d(x, z)$: $u \in B(x, z)$.

If $d(z, v) > d(x, z)$, by precircularity (considering the four points $z <_z u <_z y <_z v$) we have that $\max\{d(z, u), d(y, u)\} \leq d(y, z) \leq d(x, z)$. Moreover, considering $z <_z x <_z u <_z v$, precircularity also implies $\max\{d(z, x), d(u, x)\} \leq d(z, u) \leq d(x, z)$. Thus $u \in B(x, z)$. Since u has been arbitrarily chosen, there exists a path from y to x in $C_{<}$ of elements from $B(x, z)$.

Since the same arguments hold for an element u not in $B(x, z)$ and such that $x <_x u <_x y$, for all y in $B(x, z)$ there exists either a path from y to x in $C_{<}$ composed of elements from $B(x, z)$ or a path from y to z in $C_{<}$ of elements from $B(x, z)$. Thus $B(x, z)$ is a connected class of $C_{<}$ because there exists a path from x to z in $B(x, z)$. $\square$

We will now focus on precircularity. Because of the symmetry of 2-balls, equation (6) can be interpreted as: “there exists a path from x to z of elements of $B(x, z)$ in $C_{<}$ ”. That is, either all $t \in X$ such that $t <_z x$ are in $B(x, z)$ or all $t \in X$ such that $t <_x z$ are in $B(x, z)$.

Moreover, it is easy to show that robinsonian dissimilarities satisfy equation 6 (thus equation 5) as shown in proposition 7.

Proposition 7. Let d be a robinsonian dissimilarity admitting $<$ as compatible order. The order $<$ is then precircularly compatible with d . If $<$ is precircularly compatible with d , it is a compatible order for d if and only if:

$$d(x_1, x_n) = \max\{d(x, y) | x, y \in X\}$$

Proof. It is clear that a compatible order $<$ for a robinsonian dissimilarity d satisfies equation 6, and that for a compatible order $d(x_1, x_n) = \max\{d(x, y) | x, y \in X\}$.

Suppose that $<$ satisfies equation 6 and that $d(x_1, x_n) = \max\{d(x, y) | x, y \in X\}$. Let $x, y \in X$ such that $x < y$. If $d(x, y) = \max\{d(z, t) | z, t \in X\}$, then for all $z \in X$ such that $x < z < y$, $d(x, y) \geq \max\{d(x, z), d(y, z)\}$.

Suppose now that $d(x, y) < \max\{d(z, t) | z, t \in X\}$. If $x = x_1$ (resp. $y = x_n$), then for all $z \in X$ such that $x < z < y$, we have $x_1 < z < y < x_n$ (resp. $x_1 < x < z < x_n$) and since $d(x, y) < d(x_1, x_n)$ equation (6) ensures us that $d(x, y) \geq \max\{d(x, z), d(y, z)\}$.

If $x \neq x_1$ and $y \neq x_n$ we have $x_1 < x < z < y < x_n$. Thus, according to precircularity, either $z \in B(x, y)$ (and then $d(x, y) \geq \max\{d(x, z), d(z, y)\}$) or $\{x_1, x_n\} \subseteq B(x, y)$. In this last case, $x_n \notin B(x_1, y)$ (since $x_1 \in B(x, y)$, we have $d(x_1, y) \leq d(x, y) < d(x_1, x_n)$) and $x_1 \notin B(x, x_n)$ (for the same reason). Thus precircularity ensures us that $z \in B(x_1, y) \cap B(x, x_n)$, which leads to $d(x, y) \geq \max\{d(x, z), d(z, y)\}$ so thus the order $<$ is compatible with d . $\square$

	x	0			
	y	1	0		
	z	1	2	0	
d_1	t	3	3	1	0
	x	y	z	t	

Table 2: d satisfies equation 5 for $x \leq y \leq z \leq t$, but $\leq$ is not a compatible order for d

Note that proposition 7 no longer holds if d only satisfies equation 5 for a linear order $\leq$, as shown in table 2.

A direct consequence of proposition 7 is that if $\leq$ is a precircularly compatible order for a dissimilarity d such that there exists two consecutive elements $x_i \leq x_{i+1}$ for which $d(x_i, x_{i+1}) = \max \{d(x, y) | x, y \in X\}$, then $\leq_{x_{i+1}}$ is a compatible order for d .

We show in this last part that precircular dissimilarities are a strict subset of circular anti-robinsonian dissimilarities. Indeed, the dissimilarity d_2 of table 3 is an circular anti-robinsonian dissimilarity (the linear order $x \leq y \leq z \leq t$ holds) but not a precircular one, and proposition 8 shows that precircular dissimilarities are circular anti-robinsonian ones.

Proposition 8. *Let d be a precircular dissimilarity on X admitting $\leq$ as a circularly compatible order. Then d is also a circular anti-robinsonian dissimilarity with the same linear order.*

Proof. Let d be a circular dissimilarity, admitting $\leq$ as circularly compatible order. We label the elements of X such that $x_1 \leq x_2 \leq \dots \leq x_i \leq \dots \leq x_n$. For $1 \leq i \leq n-2$ and $i+1 < j \leq n-1$, considering $x_i \leq x_{i+1} \leq x_j \leq x_{j+1}$ and $x_{j+1} \leq x_{j+2}$ $x_i \leq x_{j+1}$ $x_{i+1} \leq x_{j+1}$ x_j , equation 6 leads to:

$$d(x_i, x_j) \geq \min \{ \max \{d(x_i, x_{i+1}), d(x_{i+1}, x_j)\}, \max \{d(x_i, x_{j+1}), d(x_{j+1}, x_j)\} \} \quad (7)$$

$$d(x_{i+1}, x_{j+1}) \geq \min \{ \max \{d(x_i, x_{j+1}), d(x_i, x_{i+1})\}, \max \{d(x_j, x_{j+1}), d(x_{i+1}, x_j)\} \} \quad (8)$$

We first show that d satisfies the two first conditions of circular anti-robinsonian dissimilarities.

The first condition supposes that $d(x_{i+1}, x_j) \leq d(x_i, x_{j+1})$. Considering the inequalities (7) and (8), $\max \{d(x_{i+1}, x_j), d(x_i, x_{i+1})\} \geq d(x_{i+1}, x_j)$ and $\max \{d(x_i, x_{j+1}), d(x_j, x_{j+1})\} \geq d(x_i, x_{j+1})$, and using (7) we get: $d(x_i, x_j) \geq d(x_{i+1}, x_j)$. By the same way, we obtain: $\max \{d(x_i, x_{i+1}), d(x_i, x_{j+1})\} \geq d(x_i, x_{j+1})$ and $\max \{d(x_{i+1}, x_j), d(x_j, x_{j+1})\} \geq d(x_{i+1}, x_j)$. Therefore, (8) leads to $d(x_{i+1}, x_{j+1}) \geq d(x_{i+1}, x_j)$.

If $d(x_{i+1}, x_j) \geq d(x_i, x_{j+1})$, by (7) we obtain $d(x_i, x_j) \geq d(x_i, x_{j+1})$. Similarly, by (8), we get $d(x_{i+1}, x_{j+1}) \geq d(x_i, x_{j+1})$.

It remains to show that the last two conditions of circular anti-robinsonian dissimilarities are satisfied.

If $d(x_{i+1}, x_n) \leq d(x_i, x_1)$, we have then $d(x_1, x_{i+1}) \geq d(x_{i+1}, x_n)$ and $d(x_i, x_n) \geq d(x_{i+1}, x_n)$ from equation 6. The third condition is satisfied. Finally, if $d(x_{i+1}, x_n) \geq d(x_i, x_1)$, we also obtain $d(x_1, x_{i+1}) \geq d(x_1, x_i)$ and $d(x_i, x_n) \geq d(x_1, x_i)$ from equation 6. The last condition is also verified $\square$

To characterize circular dissimilarities we need to add another condition, as for characterizing dissimilarities whose 2-ball hypergraph is a hypercycle. Let $\leq$ be a linear order on X , d a dissimilarity on X , and $(x, y, z, u, v) \in X^5$ such that $x \leq_x u \leq_x y \leq_x v \leq_x z$:

$$\min \{d(u, z), d(v, x)\} \leq \max \{d(x, y), d(x, z), d(y, z)\} \quad (9)$$

We say that $\leq$ is *circularly compatible* with d if and only if $\leq$ is precircularly compatible with d and satisfies equation (9). Circular compatibility is a restriction of the conditions of proposition 6. An order $\leq$ circularly compatible with d , ensures us that the 2-balls of d are connected classes of $C_{\leq}$.

It is clear that if $\leq$ is circularly compatible with d so are $\leq^{-1}$ and $\leq_x$ for all $x \in X$. Moreover the restriction of d to $Y \subseteq X$ admits as circularly compatible order the restriction from $\leq$ to Y . Conversely, if there exists $Y \subseteq X$ such that the restriction of d to Y does not admit any circularly compatible order, then d does not admit any circularly compatible order on X .

For instance, dissimilarity d_1 in figure 3 admits $t \leq x \leq y \leq z$ as circularly compatible order and dissimilarity d_2 in figure 3 does not admit any circularly compatible order.

We can now prove the equivalence between dissimilarities admitting a circularly compatible order and dissimilarities whose cluster hypergraph is a hypercycle.

Proposition 9. *Let d be a dissimilarity. The cluster hypergraph of d is a hypercycle if and only if there exists a linear order $\leq$ circularly compatible with d*

Proof. Suppose that the cluster hypergraph of d is a hypercycle. Then there exist a linear order $\leq$ such that $C_{\leq}$ is an underlying vertex cycle of it.

Let x, z be two different elements of X . $\{x, z\}$ can be extended to a maximal clique C of diameter $d(x, z)$. Since C is a connected class of $C_{\leq}$ either any y such that $x \leq_x y \leq_x z$ is in C or any t such that $z \leq_x t$ is in C . The linear order $\leq$

Conversely suppose that d and $\leq$ satisfy condition (12), and let $(x, y) \in X^2$. Since $d(x, y) \geq \min\{\max\{M(x, y, x), M(x, y, y)\}, \max\{M(y, x, x), M(y, x, y)\}\}$, one can suppose without loss of generality that $d(x, y) \geq \max\{M(x, y, x), M(x, y, y)\}$. Thus for all $z \leq_x y$ $d(x, y) \geq \max\{d(x, z), d(y, z)\}$. Hence $\leq$ is precircularly compatible with d . $\square$

Proposition 11. *Let d be a dissimilarity. A linear order $\leq$ on X is circularly compatible with d if and only if the following two conditions are satisfied:*

- (i) *The 2-ball hypergraph of d is a hypercycle admitting $C_{\leq}$ as underlying vertex cycle,*
- (ii) *for all $(x, y, u) \in X^3$ such that $x \leq_x y \leq_x u$ and $u \notin B(x, y)$:*
 - *for all $z \in B(x, y)$ such that $y \leq_x z \leq_x u$, $M(x, z, z) \leq d(x, y)$*
 - *for all $z \in B(x, y)$ such that $u \leq_x z \leq_x x$, $M(z, y, z) \leq d(x, y)$*

Proof. First, suppose that $\leq$ is circularly compatible with d . According to proposition 3, the 2-ball hypergraph of d admits $C_{\leq}$ as underlying vertex cycle. Let now be $(x, y, u) \in X^3$ such that $x \leq_x y \leq_x u$ and $u \notin B(x, y)$. Let then $z \in X$ be such that $z \in B(x, y)$ and $y \leq_x z$. There exists a maximal clique of diameter $d(x, y)$ including x, y and z . Since this clique cannot includes u :

- if $z \leq_x u$, it contains all the elements $t \in X$ such that $t \leq_x z$, which leads to $M(x, z, z) \leq d(x, y)$,
- if $u \leq_x z$, it contains all the elements $t \in X$ such that $z \leq_y t$, which leads to $M(z, y, z) \leq d(x, y)$.

Conversely, suppose that d and $\leq$ satisfy the above two conditions. The linear order $\leq$ is then precircularly compatible with d . According to proposition 9 we only have to prove that equation 9 is satisfied. Let x, y, z, u and v be elements of X such that $u \leq_x y \leq_x v \leq_x z$.

If $y \in B(x, z)$, proposition 6 ensures us that equation 9 is satisfied. Consider then that $y \notin B(x, z)$. Thus either $z \in B(x, y)$ or $x \in B(y, z)$.

If $z \in B(x, y)$, two cases may occur: either $d(x, v) \leq d(x, y)$ (and equation 9 is satisfied) or $d(x, v) > d(x, y)$ (thus $v \notin B(x, y)$). In the second case, condition (ii) ensures us that $M(z, y, z) \leq d(x, y)$: thus $d(z, u) \leq d(y, z) \leq d(x, y) \leq \max\{d(x, y), d(x, z), d(y, z)\}$ and equation 9 is satisfied.

If $x \in B(y, z)$, since $v \leq_y z \leq_y x \leq_y u$, if $u \in B(y, z)$ equation 9 is satisfied, or, by $d(u, z) \leq d(y, z)$, condition (ii) ensures us that $M(y, x, x) \leq d(y, z)$ thus $d(x, v) \leq d(y, z) \leq \max\{d(x, y), d(x, z), d(y, z)\}$ and equation 9 is also satisfied.

Then the order $\leq$ is precircularly compatible with d and equation 9 is satisfied: $\leq$ is circularly compatible with d . $\square$ $\square$

Proposition 10 leads to a way to recognize if a given dissimilarity d is circularly compatible with a linear order $\leq$ in $\mathcal{O}(|X|^3)$ time and $\mathcal{O}(|X|^2)$ space.

Indeed:

- computing and storing $M(x, y, y)$ and $M(y, x, x)$ for all $(x, y) \in X^2$ can be done in $\mathcal{O}(|X|^3)$ time and $\mathcal{O}(|X|^2)$ space,
- checking if the set of 2-balls of d admits $C_{\leq}$ as underlying vertex circle can be done in $\mathcal{O}(|X|^3)$ time and $\mathcal{O}(|X|^2)$ space,
- checking condition (ii) of proposition 10 can also be done in $\mathcal{O}(|X|^3)$ time and $\mathcal{O}(|X|^2)$ space if the $M(x, z, z)$ and $M(z, y, z)$ have been previously stored.

Since there is at most $\mathcal{O}(|X|^2)$ balls or 2-balls for a given dissimilarity, it is obvious that checking if the ball or the 2-ball hypergraph of a given dissimilarity admits a given circle as underlying vertex circle may also be done in $\mathcal{O}(|X|^3)$ time and $\mathcal{O}(|X|^2)$ space.

The next section shows that this result can be extended to check if a given dissimilarity is circular or not.

5.2 Recognizing if a given dissimilarity is circular

Section 3.2 gives a polynomial algorithm to check whether a hypergraph $\mathcal{H} = (X, \mathcal{E})$ is a hypercycle or not. This algorithm takes $\mathcal{O}(|X| * |\mathcal{E}|)$ time and $\mathcal{O}(|X| * |\mathcal{E}|)$ space to be performed.

When dealing with dissimilarities, the space taken by this algorithm can be reduced from $\mathcal{O}(|X||\mathcal{E}|)$ to $\mathcal{O}(|X|^2)$. Indeed, at each step of the algorithm we only consider one cluster, this cluster can then be constructed during this iteration. Thus, if the algorithm of section 3.2 iterates on all the possible balls, 2-balls or 3-balls, it takes only $\mathcal{O}(|X|^2)$ memory space and $\mathcal{O}(m|X|)$ time where m is the number of possible clusters ($m = |X|^2$ for balls and 2-balls and $m = |X|^3$ for 3-balls).

This (modified) algorithm allows then to recognize if the 2-ball set of a given dissimilarity d admits an underlying vertex cycle or not in $\mathcal{O}(|X|^3)$ operation and in $\mathcal{O}(|X|^2)$ time. Moreover, if the 2-ball set of d is a hypercycle, it gives one of its underlying vertex cycle.

At first glance, recognizing if a given dissimilarity d is circular or not takes $\mathcal{O}(|X|^4)$ operations (there are $\mathcal{O}(|X|^3)$ possible 3-balls). But proposition 12 gives a way to lower this complexity. Indeed, it is sufficient to verify that the 2-balls hypergraph of d is a hypercycle (which can be done in $\mathcal{O}(|X|^3)$ operations) and check if an order associated to this cycle is circular or not.

Proposition 12. *If d is a circular dissimilarity, any order $\leq$ such that the 2-ball hypergraph of d is rigid on $C_{\leq}$ is circularly compatible with d .*

Proof. Let d be a circular dissimilarity. The 2-ball hypergraph of d is then also a hypercycle (proposition 4). Let $<$ be a linear order such that $C_{<}$ is an underlying vertex cycle of its 2-ball hypergraph.

We suppose that $<$ is not circularly compatible with d , and let A be a cluster of d which is not a connected class of $C_{<}$. There exist then $t_1, t_2 \in A$ and $z, u \notin A$ such that $z <_{t_1} t_2 <_{t_1} u$. We note A_1 the elements x of A such that $x <_u z$ and A_2 the elements y of A such that $y <_z u$ (as shown in figure 4). It is clear that neither A_1 nor A_2 are empty, that $A_1 \cap A_2 = \emptyset$ and $A_1 \cup A_2 = A$.

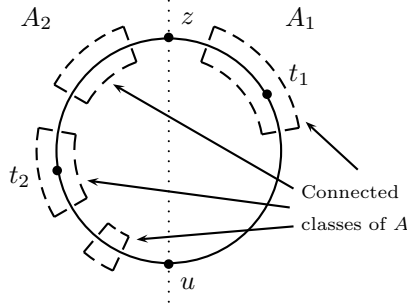


Figure 4: The cluster A

Since A is a cluster of d , there exists $(x, y) \in A^2$ such that $d(x, z) > \text{diam}_d(A)$ and $d(y, u) > \text{diam}_d(A)$. Three cases are then possible:

Case 1: $x \in A_1$ (resp. in A_2) and $y \in A_2$ (resp. in A_1). The 2-ball $B(x, y)$ cannot be a connected part of $C_{<}$ because neither z nor u are in $B(x, y)$ (note that $d(x, y) \leq \text{diam}_d(A)$). This case is not possible.

Case 2: $x = y$. According to proposition 3, the ball $B(x, \text{diam}(A))$ is then a connected part of $C_{<}$. But this cannot be true because neither z nor u are in $B(x, \text{diam}(A))$ even though t_1 and t_2 are in $B(x, \text{diam}(A))$. This case is also not possible.

Case 3: If case 1 and case 2 are not satisfied, x and y are then necessarily in A_1 (resp. in A_2). Moreover, $d(x, u) \leq \text{diam}_d(A)$ and $d(y, t) \leq \text{diam}_d(A)$.

Since case 1 and case 2 are not satisfied, we also have that $d(t_2, z) \leq \text{diam}_d(A)$ and $d(t_2, u) \leq \text{diam}_d(A)$ (consider t_1 instead of t_2 if x and y are in A_2). Moreover, since $B(x, y)$ is a connected part of $C_{<}$ $d(x, y) < \max\{d(x, t_2), d(y, t_2)\}$. Thus:

- $\min\{d(x, z), d(u, y)\} > \max\{d(x, y), d(x, t_2), d(y, t_2), d(z, y), d(z, t_2), d(u, x), d(u, t_2)\}$
- $d(x, y) < \max\{d(x, t_2), d(y, t_2)\}$

Thus, since d is a circular dissimilarity on X , d restricted to $\{x, y, z, t, u\}$ is also a circular dissimilarity. But, according to the above conditions, the clusters of d restricted to $\{x, y, z, t, u\}$ contain the clusters of figure 3 (a), which are not a hypercycle: d can't be a circular dissimilarity.

Finally, the three cases are not possible: A is a connected part of $C_{<}$, which concludes the proof $\square$

Recognizing if a given dissimilarity d is circular or not can then be done in $\mathcal{O}(|X|^3)$ time and $\mathcal{O}(|X|^2)$ space by:

1. checking if the 2-ball set of d is a hypercycle or not (using the (modified) algorithm of section 3.2),
2. if C is an underlying vertex cycle of the 2-ball set of d , let $<$ be a linear order such that $C = C_{<}$,
3. checking if $<$ is circularly compatible with d (section 5.1).

6 Bijection Theorem

This section will give a characterization of indexed hypercycles which are cluster hypergraphs of some dissimilarity. We will give a bijection between some indexed hypergraphs and circular dissimilarities.

Unlike interval hypergraphs and robinsonian dissimilarities, there exist hypercycles that cannot be associated to any dissimilarities. For instance, the hypercycle of figure 2 is not a cluster hypergraph of some dissimilarity.

To achieve this task we need to introduce some definitions. A *clustering system* $\mathcal{H} = (X, \mathcal{E})$ is a hypergraph such that:

- $X \in \mathcal{E}$,
- $\{x\} \in \mathcal{E}$ for all $x \in X$.

Since a (proper) dissimilarity always has the singletons and X as cluster, it is obvious that hypercycles that are cluster hypergraphs of some dissimilarities are clustering systems.

Each clustering system $\mathcal{H} = (X, \mathcal{E})$ (and more generally, each hypergraph) can be valued by a map f from $\mathcal{E}$ to $\mathbb{R}^+$. A *valuated clustering system* is then a triple $(X, \mathcal{E}, f)$ where $\mathcal{H} = (X, \mathcal{E})$ is a clustering system and f a map from $\mathcal{E}$ to $\mathbb{R}^+$.

A valuated clustering system $(X, \mathcal{E}, f)$ is said to be:

- *pre-indexed* if f is isotone:

$$\forall A, B \in \mathcal{E}, A \subseteq B \Rightarrow f(A) \leq f(B)$$

- *indexed* if f is strictly isotone:

$$\forall A, B \in \mathcal{E}, A \subsetneq B \Rightarrow f(A) < f(B)$$

Given a pre-indexed clustering system $(X, \mathcal{E}, f)$, we can define the map f^* from 2^X to $\mathbb{R}^+$ such that:

$$f^*(A) = \min\{f(B) | B \in \mathcal{E}, A \subseteq B\}$$

To characterize hypercycles that are cluster hypergraphs, we have to restrict the general bijection between cluster hypergraphs of dissimilarities and valuated clustering systems (Batbedat, 1988 and Bertrand, 2000).

The general bijection theorem between dissimilarities and valuated clustering system is then:

Theorem 1 (Bertrand, 2000). *There is a bijection between the set of all (proper) dissimilarities on X and the set of all the indexed clustering systems $(X, \mathcal{E}, f)$ such that:*

- $f^{-1}(0) = \{\{x\} | x \in X\}$
- for all $(A, B, C) \in \mathcal{E}^3$,

$$f^*((A \cap B) \cup (A \cap C) \cup (B \cap C)) \leq \max\{f(A), f(B), f(C)\} \quad (13)$$

$\mathcal{H} = (X, \mathcal{E})$ is then equal to the cluster hypergraph $\mathcal{H}_d = (X, \mathcal{E}_d)$ of some dissimilarity d and $f = \text{diam}_d$.

A particular case of clustering system are *weak hierarchies* (Bandelt and Dress, 1989). A weak hierarchy is a hypergraph $\mathcal{H} = (X, \mathcal{E})$ such that for all $(A, B, C) \in \mathcal{E}^3$, $A \cap B \cap C \in \{A \cap B, B \cap C, A \cap C\}$. One of their interests is that for any $(A, B, C) \in \mathcal{E}^3$, $(A \cap B) \cup (A \cap C) \cup (B \cap C)$ is a subset of A , B , or C . Thus, every indexed weak hierarchy satisfies the second condition of theorem 1.

We will say that a triple $(A, B, C) \in \mathcal{E}^3$ is *weakly hierarchical* for the clustering system $\mathcal{H} = (X, \mathcal{E})$ if $A \cap B \cap C \in \{A \cap B, B \cap C, A \cap C\}$. According to this definition, one can define an indexed clustering system $(\mathcal{H} = (X, \mathcal{E}), f)$ as an *indexed circular pseudo-hierarchy* if and only if:

- $f^{-1}(0) = \{\{x\} | x \in X\}$
- there exists a linear order $<$ circularly compatible with $\mathcal{H}$,
- for any $A, B, C \in \mathcal{E}$, one of the following two conditions is always satisfied:
 - (A, B, C) is weakly hierarchical,
 - $\min\{f^*(A \cup B), f^*(A \cup C), f^*(B \cup C)\} \leq \max\{f(A), f(B), f(C)\}$

Theorem 2 shows the bijection between indexed circular pseudo-hierarchy and circular dissimilarities.

Theorem 2. *Let $(X, \mathcal{E}, f)$ be an indexed circular pseudo-hierarchy admitting $<$ as circularly compatible order. There exists then an unique circular dissimilarity d such that $(X, \mathcal{E}_d, \text{diam}_d) = (X, \mathcal{E}, f)$ and for any given circular dissimilarity d with compatible order $<$, there exists an unique circular pseudo hierarchy equal to $(X, \mathcal{E}_d, \text{diam}_d)$*

Proof. According to theorem 1, there exists an unique dissimilarity d on X such that $(X, \mathcal{E}, f) = (X, \mathcal{E}_d, \text{diam}_d)$ if and only if

- $f^{-1}(0) = \{\{x\} | x \in X\}$
- for all $A, B, C \in \mathcal{E}$,

$$f^*((A \cap B) \cup (A \cap C) \cup (B \cap C)) \leq \max\{f(A), f(B), f(C)\}$$

If $A \cap B \cap C \in \{A \cap B, B \cap C, A \cap C\}$, we can suppose without loss of generality that $A \cap B \cap C = A \cap B$. Thus, $(A \cap B) \cup (A \cap C) \cup (B \cap C) \subseteq C$ and since f is an index on $\mathcal{H}$, $f^*((A \cap B) \cup (A \cap C) \cup (B \cap C)) \leq f(C) \leq \max\{f(A), f(B), f(C)\}$.

If $A \cap B \cap C \notin \{A \cap B, B \cap C, A \cap C\}$, $(A \cap B) \cup (A \cap C) \cup (B \cap C)$ is a nonconnected class of $C_{<}$, each cluster D of $\mathcal{E}_d$ such that $(A \cap B) \cup (A \cap C) \cup (B \cap C) \subseteq D$ necessarily includes $A \cup B$, $A \cup C$, or $B \cup C$. Theorem 1 is then equivalent for hypercycles to the conditions of theorem 2, which concludes the proof. $\square$

Theorem 2 is the restriction to indexed circular pseudo hierarchies of the general bijection (theorem 1). This restriction allows the general one to be simplified by introducing weakly hierarchical triple.

7 Approximations

Since a given dissimilarity on X has little chance to be circularly compatible with a linear order, we propose in this section an algorithm in $\mathcal{O}(n^4)$ operations to approximate a given dissimilarity on X with a circularly compatible dissimilarity on a given linear order $<$.

This algorithm can be easily extended, using general optimization techniques, to “find” a good linear order.

7.1 Approximation at fixed linear order

To reduce the complexity of the algorithm, we will focus on circular dissimilarities whose cluster hypergraph is closed under finite intersection. Those dissimilarities are called *quasi-ultrametrics* (for more details about quasi-ultrametrics, one can see Bandelt and Dress 1989, Bandelt 1992, Diatta and Fichet 1994, Diatta 1996 or Brucker 2001).

A dissimilarity d on X is a quasi-ultrametric if and only if it satisfies the so called *four point condition*: for all $x, y, z, t \in X$, $\max\{d(z, x), d(z, y)\} \leq d(x, y) \Rightarrow d(z, t) \leq \max\{d(t, x), d(t, y), d(x, y)\}$.

The main interest using quasi-ultrametrics is that their cluster hypergraph and 2-ball hypergraph coincide. So, a quasi-ultrametric is a circular one if and only if only the two first conditions of definition are satisfied (in other words, a quasi-ultrametric is circular if and only if its 2-ball hypergraph is a hypercycle). Proposition 13 states this.

Proposition 13. *A quasi-ultrametric d on X admits a linear order $\leq$ as circularly compatible order if and only if for all $x, y \in X$, the 2-ball $B(x, y)$ contains either the interval $[xy]$ for the linear order $\leq_x$, or the interval $[yx]$ for the linear order $\leq_y$.*

Proof. Let d be a quasi-ultrametric on X admitting $\leq$ as circularly compatible order. For $x, y \in X$, $B(x, y)$ is then a connected class of the circle $C_{\leq}$ and thus contains either the interval $[xy]$ for the linear order $\leq_x$, or the interval $[yx]$ for the linear order $\leq_y$.

Conversely, suppose that d is a quasi-ultrametric on X such that for all $x, y \in X$, the 2-ball $B(x, y)$ contains either the interval $[xy]$ for the linear order $\leq_x$, or the interval $[yx]$ for the linear order $\leq_y$. Let $B(x, y)$ be a 2-ball of d and $z, t \in B(x, y)$. Diatta and Fichet 1984 proved that we have $B(z, t) \subseteq B(x, y)$. Thus, since the interval $[zt]$ for the linear order $\leq_z$, or the interval $[tz]$ for the linear order $\leq_t$ is included in $B(z, t)$, it is also included in $B(x, y)$. This leads to the fact that a path from z to t in the cycle $C_{\leq}$ is included in $B(x, y)$. Thus, as z and t are any elements of $B(x, y)$, $B(x, y)$ is a connected class of $C_{\leq}$. The 2-ball hypergraph of d is a hypercycle admitting $C_{\leq}$ as underlying vertex cycle, and since the 2-ball hypergraph and the cluster hypergraph of d coincide, d is a circular dissimilarity admitting $\leq$ as circularly compatible order $\square$

To ensure the result of our algorithm to be a quasi-ultrametric, we will use notions developped and proven in Brucker (2001, 2005). We call a *quatuor* of X any four element subset of X . For x and y in X , $Q[x, y]$ denotes the set of all quatuors that contain x and y . The quatuor Q is said to be *quasi-ultrametric* for a dissimilarity d if and only if the restriction of d to Q is a quasi-ultrametric. It is clear that d is a quasi-ultrametric if and only if all the quatuors are quasi-ultrametric for d . The following proposition is demonstrated in Brucker 2001 and is the basis of the further algorithm.

Proposition 14. *Let d be a dissimilarity on X and Q a quatuor:*

- *if the two largest values of d on Q are equal, then Q is a quasi-ultrametric quatuor for d ,*

Let us consider the complete graph $K_X = (X, E_X)$, valued by d . If Q is a quatuor, we note $K_Q = (Q, E_Q)$ the complete graph valued by d . Let Q be a quatuor of X and F a subset of E_X . We note $\delta(Q, F)$ the number of edges from K_Q that are in F (then $0 \leq \delta(Q, F) \leq 6$).

Let $\leq$ be a linear order on X . We shall construct by induction a sequence E_k of subsets from E_X and a sequence d_i of dissimilarities on X in order to transform d into an circular quasi-ultrametric admitting $\leq$ as circularly compatible order:

1. $E'_0 := \emptyset$, $d'_0 := d$;
2. E'_k and d'_k being constructed, we consider an edge $xy \notin E'_k$ such that $d'_k(x, y)$ is a minimum for all edges that are not in E'_k (we say that xy is *examined*);
 - (a) $d'_{k+1} := d'_k$
 - (b) If both paths from x to y in $C_{\leq}$ are not composed of elements of the 2-ball $B(x, y)$ for d'_k , choose one and for all z in this path, if $xz \notin E'_k$ then $d'_{k+1}(x, z) = d'_k(x, y)$, and if $yz \notin E'_k$ then $d'_{k+1}(y, z) = d'_k(x, y)$.
 - (c) For all quatuors Q from $Q[x, y]$ that are not quasi-ultrametric for d'_k and such that $\delta(Q, E'_k) = 4$, we note $u_Q v_Q$ the unique edge such that $d'_k(u_Q, v_Q) = \text{diam}_{d'_k}(Q)$. We then have $d'_k(u_Q, v_Q) > d'_k(x, y)$; this value is greater than the four remaining values for d'_k . We then set $d'_{k+1}(u_Q, v_Q) = d'_k(x, y)$.
 - (d) $E'_{k+1} = E'_k \cup xy$

Since $d'_{n(n-1)/2}$ is the output of the algorithm, we note: $q_{\leq}(d) = d'_{n(n-1)/2}$.

The following proposition will then prove that $q_{\leq}(d)$ is an circular quasi-ultrametric and that $C_{\leq}$ is one of its underlying vertex cycle. Moreover the construction of this approximation may be computed in $\mathcal{O}(n^4)$ operations.

Proposition 15. *The sequences $(E'_k)_{1 \leq k \leq n(n-1)/2}$ and $(d'_k)_{1 \leq k \leq n(n-1)/2}$ have the following properties:*

1. For $k < \frac{n(n-1)}{2}$, $E'_k \subsetneq E'_{k+1}$, and $d'_{k+1} \leq d'_k$,
2. For all edges $xy \in E$, each quatuor Q from $Q[x, y]$ such that $\delta(Q, E'_k) \geq 5$ is quasi-ultrametric for d'_k ,
3. For all edges $xy \in E'_k$, one path from x to y in $C_{\leq}$ is in the 2-ball $B_{d'_k}(x, y)$.

Proof. Assertion 1 is clear.

We use the notations of the construction of $q_{\leq}(d)$. We show by induction on step k of the construction that:

- (i) $\max\{d'_{k+1}(x, y) | xy \in E'_{k+1}\} \leq \min\{d'_{k+1}(x, y) | xy \in E \setminus E'_{k+1}\}$
- (ii) $d'_k(x, y) = d'_{k+1}(x, y)$ for any edge xy in E'_{k+1}

For $k = 0$, the two properties are clearly satisfied because $d'_1 = d'_0 = d$. We suppose that the two properties are verified at step $k - 1$. Let $x^*y^* \in E \setminus E'_k$ be the edge examined at step k of the construction ($d'_k(x^*, y^*) = \min\{d'_k(x, y) | xy \in E \setminus E'_k\}$).

Modifications of dissimilarity d'_k are made in step (2b) and (2c) of the construction. For step (2b), let z be in the chosen path from x^* to y^* in $C_{\leq}$. If $x^*z \notin E'_k$ (resp. $y^*z \notin E'_k$), we have $d(x^*, z) \geq d(x^*, y^*)$ (resp. $d(y^*, z) \geq d(x^*, y^*)$) by induction hypothesis. For step (2c), let $Q \in Q[x^*, y^*]$ be a non quasi-ultrametrical quatuor for d'_k such that $\delta(Q, E'_k) = 4$. Then there exists a unique edge $u_Q v_Q$ from Q such that $d'_k(u_Q, v_Q) = \text{diam}_{d'_k}(Q)$. Since $d'_k(u_Q, v_Q) > d'_k(x^*, y^*)$, $u_Q v_Q$ is in $E \setminus E'_k$. We then have $d'_{k+1}(u_Q, v_Q) = d'_k(x^*, y^*) = d'_{k+1}(x^*, y^*)$. Thus, each edge xy such that $d'_{k+1}(x, y) < d'_k(x, y)$ fulfills:

- $xy \in E \setminus E'_{k+1}$,
- $d'_{k+1}(x, y) = d'_{k+1}(x^*, y^*)$.

Since $E'_{k+1} = E'_k \cup \{x^*y^*\}$, assertions (i) and (ii) are proved.

Thus, for all $0 \leq i \leq j$:

- $d'_i(x, y) = d'_j(x, y)$ for all xy in E'_i ,
- $\max\{d'_j(x, y) | xy \in E'_i\} \leq \min\{d'_j(x, y) | xy \in E \setminus E'_i\}$

These two properties justify the construction of the sequence (d'_k) of dissimilarities, and prove assertions 2 and 3 $\square \quad \square$

Table 4 shows an example of a dissimilarity d approximated by $q_{\leq}(d)$ for the linear order $x < y < z < t < u$.

$d :$	x	0				
	y	1	0			
	z	3	1	0		
	t	3	1	3	0	
	u	2	1	3	1	0
	x	y	z	t	u	

$d_2 :$	x	0				
	y	1	0			
	z	3	1	0		
	t	3	1	1	0	
	u	2	1	1	1	0
	x	y	z	t	u	

Table 4: d and $q_{\leq}(d)$ for the linear order $x < y < z < t < u$

7.2 Global approximation

To find an “optimal” order, we proceed by 2-optimisation on a given order, using algorithm described in section 7.1. After finding an approximation of the dissimilarity d on a given order, we explore the neighbours of this order in the sense of transpositions, by swapping two elements. Thus, an order has $\frac{n(n-1)}{2}$ neighbours. We consider dissimilarities as vectors in a $\frac{n(n-1)}{2}$ -dimensional vector space and we use the L_2 distance between the original dissimilarity d and the circular dissimilarity δ within this space to obtain a quality measure:

$$\|d - \delta\|_2^2 = \sum_{i=1}^n \sum_{j=i+1}^n (d(x_i, x_j) - \delta(x_i, x_j))^2 \quad (14)$$

The algorithm consists of:

1. Taking an order $\leq$ on X ,
2. Approximating d by a circular dissimilarity δ compatible with $\leq$,
3. Taking x and y in X . Exchange x and y within $\leq$ to build a circular order $\leq'$,
4. Approximating d by a circular dissimilarity δ' compatible with $\leq'$,
5. If $\|d - \delta\|_2 > \|d - \delta'\|_2$ then replacing $\leq$ by $\leq'$,
6. Going back to phase 3 unless all neighbours of $\leq$ have been tested and rejected.

Dear	0	19.9	29.5	29.5	29.5	29.5	29.5	29.5	29.5	29.5	29.5	22.6
Rabbit	19.9	0	20.3	20.3	20.3	20.3	20.3	20.3	29.5	29.5	29.5	22.6
Pig	41.1	25.9	0	17.0	17.0	17.0	17.0	17.0	29.3	29.3	29.5	29.5
Bear	40.1	20.3	21.4	0	16.1	16.1	16.1	16.1	22.6	22.6	24.1	29.5
Cow	43.6	27.9	17.0	27.7	0	07.7	08.2	08.2	22.6	22.6	24.1	29.5
Horse	43.0	30.1	25.7	29.0	07.7	0	07.2	07.2	22.6	22.6	24.1	29.5
Sheep	53.1	32.5	19.6	16.1	08.2	15.2	0	01.8	22.6	22.6	24.1	29.5
Goat	44.5	21.8	19.5	19.1	11.0	07.2	01.8	0	17.0	17.0	24.1	29.5
Lion	44.7	33.3	29.3	22.6	24.5	28.6	35.0	23.1	0	17.0	24.1	29.5
Dog	50.9	39.4	45.0	49.6	30.2	24.0	46.8	17.0	26.9	0	02.0	27.5
Cat	56.1	41.5	43.0	47.2	30.9	25.3	47.1	29.0	24.1	02.0	0	24.8
Mouse	33.9	22.6	34.9	29.5	34.1	32.6	51.9	39.6	33.2	27.5	24.8	0

Table 5: Henley data: dissimilarity d_H (down) and its circular approximation (up)

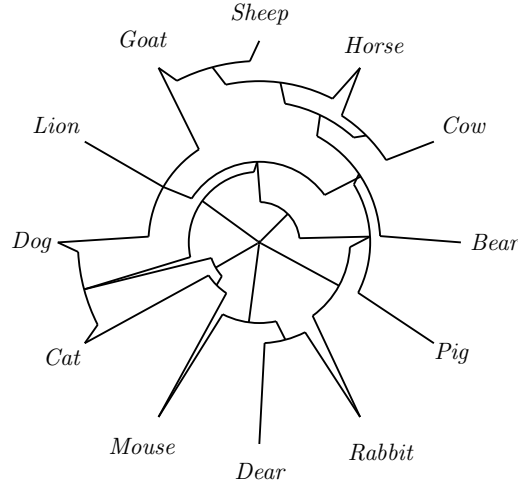


Figure 5: Approximation of d_H by a circular quasi-ultrametric

8 Application

Henley data estimates the perceived proximities between 12 types of animals: twenty-one subjects were asked to write a list of animals. Twelve were common to all those lists; in each list Henley 1969 counted the number of entries separating each pair. Table 5 (down) is obtained by taking the average of all those distances, reduced to a common scale. These data were widely studied, see for instance Barthélemy and Guénoche 1988, Abdi Barthélemy and Luong 1984 and Abdi 1990.

Those data are the typical case where circular dissimilarities approximation gives better results than robinsonian one. Indeed those data are not linked with a particular linear order, thus there is no reason for given to two elements a particular role (the first and the last element of the compatible order).

Using the algorithm described in section 7.1 within a simulated annealing optimisation method we obtain the following circular quasi-ultrametric δ_H . There is an unique cycle compatible with this dissimilarity. Table 5 represents d_H on this particular cycle, and the circular quasi-ultrametric corresponding to the circular pyramid of the figure 5.

Figure 6 shows the approximations with the average linkage (left) and the Durand (1989) algorithm (right). We see that the clusters found in the hierarchy are conserved for both the pyramidal and the circular representation and that the clusters found with the pyramidal representation sounds compatible with the circular one. Nevertheless, the circular representation is richer for the sheep and the mouse (the first and the last element of the linear order).

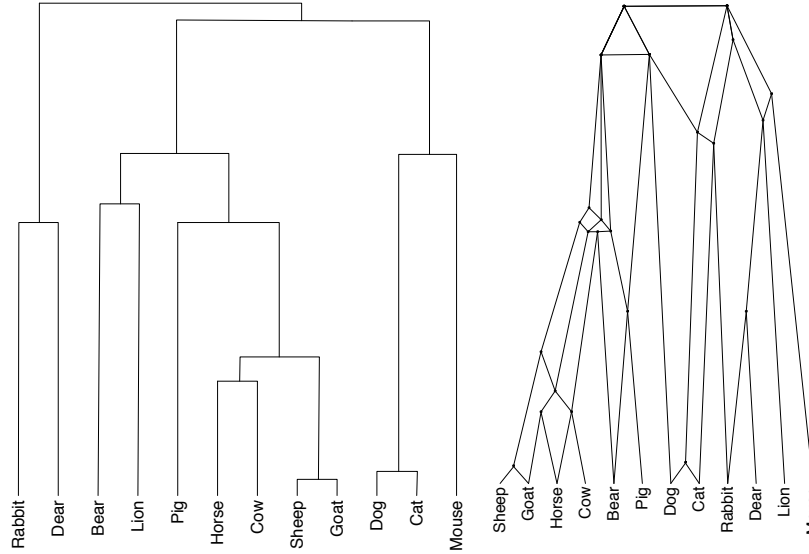


Figure 6: Approximation of d_H by an ultrametric (left) and robinsonian dissimilarity (right)

9 Conclusion

Hypercycles and circular dissimilarities (or precircular dissimilarities) can be seen as a natural extension of interval hypergraphs and robinsonian dissimilarities. As interval hypergraphs, they can be easily be graphically represented (see section 8).

Moreover, recognition of circular (or precircular) dissimilarities, or hypercycles are, from a combinatorial point of vue, not bigger than for robinsonian dissimilarities (it takes $\mathcal{O}(n^3)$ operations and $\mathcal{O}(n^2)$ space to recognize if a given dissimilarity is robinsonian, Chepoi and Fichet 1997) and the algorithm can, in combination with proposition 1, be used to recognize either circular or robinsonian dissimilarities.

Finally, circular dissimilarities and hypercycle seem to be a good classification tool because they do not induce, as robinsonian dissimilarities and interval hypergraph, a special treatment of the first and the last element of the associated linear order.

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